

# Mediapipe Palm Follow (No-PTZ)

## 1. Experiment Objective

Use MediaPipe to detect the palm, with body rotation tracking + distance following keeping the palm centered in the frame and maintaining a constant distance.

## 2. Experiment Principle

### 1. Key Terms

| Term                       | Description                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------|
| MediaPipe Hands            | Google's lightweight hand keypoint detection that outputs 21 hand skeleton points                    |
| Palm center (5th keypoint) | The MCP joint at the base of the index finger (index 5), used as the reference center for following  |
| Distance Follow            | Estimates distance from the palm bounding box size, with PD control maintaining the desired distance |
| Hand Skeleton              | 21 keypoints connected according to the hand topology, visualized in real time                       |
| QoS                        | best_effort (image input + cmd_vel downlink), depth=1                                                |

### 2. Technical Architecture

```
/camera/image_raw (BEST_EFFORT)
  → hand_tracker_node (enable_distance_follow:=true)
    └─ MediaPipe Hands → 21 keypoints → palm center + bounding box size
    └─ PD angular velocity + bounding box size distance following → /cmd_vel
```

A MediaPipe Python environment is required (for both ROS2 and the workspace environment).

## 3. Core Topics Quick Reference

### Firmware Downlink

| Topic    | Message Type        | QoS                        | Field     | Range            | Direction                                |
|----------|---------------------|----------------------------|-----------|------------------|------------------------------------------|
| /cmd_vel | geometry_msgs/Twist | best_effort + keep_last(1) | linear.x  | -0.20~0.20 m/s   | <div>+ = forward, - = backward</div>     |
|          |                     |                            | angular.z | -0.60~0.60 rad/s | <div>+ = left turn, - = right turn</div> |

### 3. Experiment Steps

#### 1. Hardware Preparation

| Hardware             | Requirement                       |
|----------------------|-----------------------------------|
| PTZ Version          | ✗                                 |
| No-PTZ Version       | ☑                                 |
| Required Peripherals | ESP32 camera (mounted on bracket) |

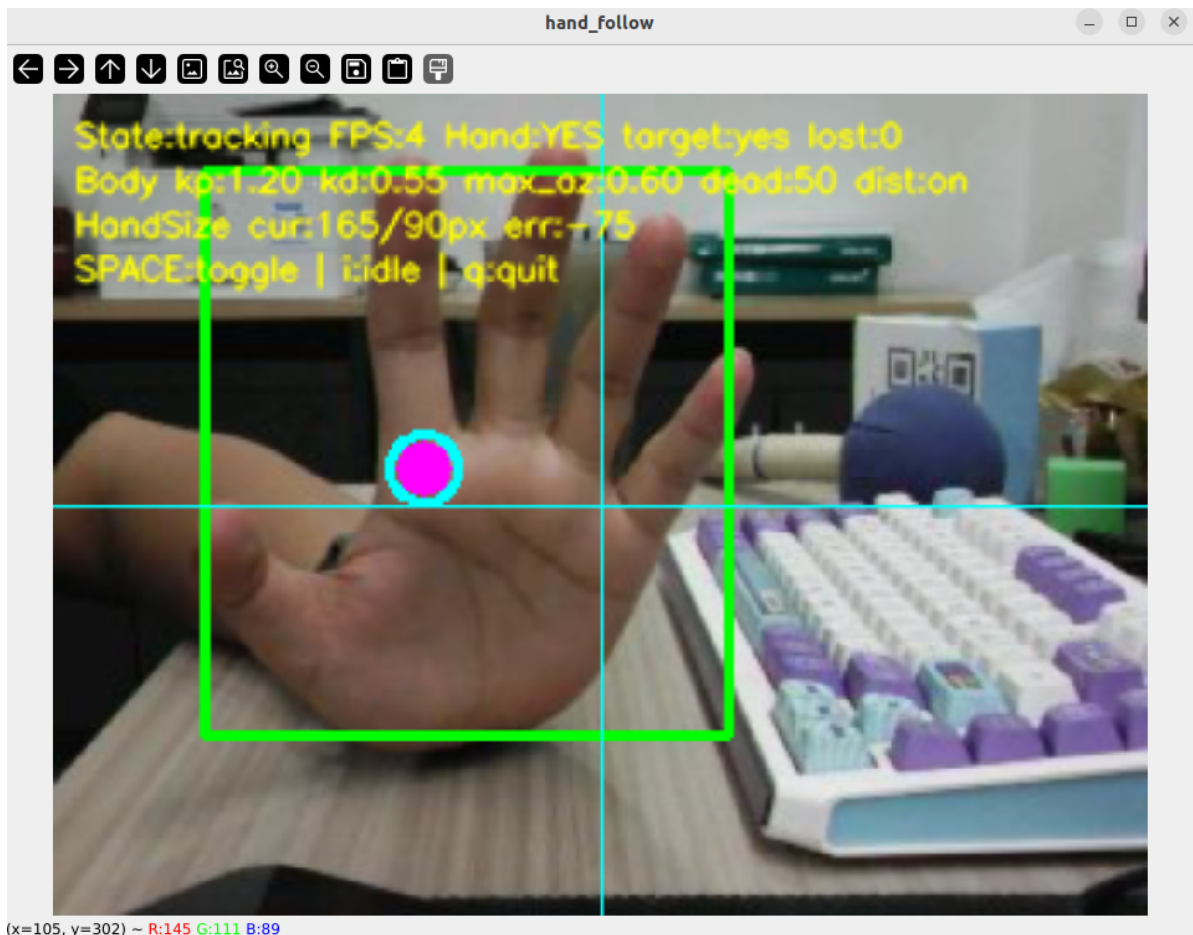
#### 2. Start Agent + Camera + Joint Model (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```

```
yahboom@yahboom: $ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-15-04-46-269642-yahboom-579370
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [579411]
[INFO] [camera_driver-2]: process started with pid [579413]
[INFO] [robot_state_publisher-3]: process started with pid [579415]
[INFO] [joint_state_publisher-4]: process started with pid [579417]
[INFO] [ekf_node-5]: process started with pid [579419]
[micro_ros_agent-1] [1785222287.404236] info | UDPv4AgentLinux.cpp | init | running... | port: 8090
[robot_state_publisher-3] [INFO] [1785222288.847816851] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785222288.848013042] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785222288.848028942] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848037122] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785222288.848043874] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785222288.848050538] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785222288.848057264] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785222288.848063747] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785222288.848070240] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848076888] [robot_state_publisher]: got segment rfwheel_Link
[joint_state_publisher-4] [INFO] [1785222289.372349515] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[camera_driver-2] [INFO] [1785222289.716161193] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[camera_driver-2] [WARN] [1785222292.705741414] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785222294.072701944] [esp32_ip_camera_publisher]: IP camera stream opened successfully
```

#### 3. Start Palm Follow (Terminal 2)

```
ros2 launch microros_v2_no_ptz_visual_control_pkg hand_follow.launch.py
```



## 6. Operation Instructions

| Key     | Function              |
|---------|-----------------------|
| Space   | Toggle follow/standby |
| q / ESC | Exit                  |

**Distance Follow Behavior:** Palm approaches (bounding box grows) → move backward; moves away (bounding box shrinks) → move forward; size within tolerance → stay still.

## 4. Experiment Result

- Palm approaches → the car moves backward to keep the distance
- Palm moves away → the car moves forward to follow
- 21 keypoints and skeleton connections displayed in real time
- Palm moves out of the frame → stops moving

## 5. Code Explanation

Source code paths:

- Launch:  
`~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/launch/hand_follow.launch.py`
- Node source:  
`~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/hand_tracker_node.py`

- Hand detector:  
`~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/hand_detector.py`

## 1. Core Node Explanation

`hand_follow.launch.py` launches `hand_tracker_node`, with `enable_distance_follow` defaulting to `true`.

**Distance Following:** The bounding box size is computed from the 21 keypoints (the larger of  $\max(x)-\min(x)$  and  $\max(y)-\min(y)$ ) and compared with the desired size `target_hand_size_px` (90 px). Once the deviation exceeds the `distance_tolerance_hand_size_px` (12 px) deadband, the PD computes the linear.x output.

## 2. Key Parameters

Parameter definition files:

- Launch:  
`~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/launch/hand_follow.launch.py`
- Node:  
`~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/hand_tracker_node.py`

| Parameter                                           | Type  | Default                               | Description                       |
|-----------------------------------------------------|-------|---------------------------------------|-----------------------------------|
| <code>deadband_x_pixel</code>                       | float | <code>50.0</code> px                  | Horizontal deadband               |
| <code>angular_kp</code> / <code>angular_kd</code>   | float | <code>1.2</code> / <code>0.55</code>  | Angle PD gains                    |
| <code>max_angular_z</code>                          | float | <code>0.6</code> rad/s                | Maximum turning speed             |
| <code>enable_distance_follow</code>                 | bool  | <code>true</code>                     | Enable distance following         |
| <code>target_hand_size_px</code>                    | float | <code>90.0</code> px                  | Desired palm bounding box size    |
| <code>distance_tolerance_hand_size_px</code>        | float | <code>12.0</code> px                  | Distance tolerance deadband       |
| <code>distance_kp</code> / <code>distance_kd</code> | float | <code>0.05</code> / <code>0.03</code> | Distance PD gains                 |
| <code>max_linear_x</code>                           | float | <code>0.2</code> m/s                  | Maximum forward/backward speed    |
| <code>lost_frame_threshold</code>                   | int   | <code>5</code>                        | Lost-target tolerance frame count |

### 3. Node Description

| Node                           | Package                                            | Description                                                 |
|--------------------------------|----------------------------------------------------|-------------------------------------------------------------|
| <code>hand_tracker_node</code> | <code>microros_v2_no_ptz_visual_control_pkg</code> | MediaPipe palm tracking + following, 21 keypoints + dual PD |

### 6. Common Issues

| Issue                        | Cause                                 | Solution                              |
|------------------------------|---------------------------------------|---------------------------------------|
| Frequent palm target loss    | Palm turned over or moving quickly    | Keep the palm plane facing the camera |
| Distance following overshoot | PD gains too large                    | Lower <code>distance_kp</code>        |
| Palm not detected            | Too far away or insufficient lighting | Move closer to the camera             |